Improving LLM Bargaining Ability via Self-Play Reinforcement Learning

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ABSTRACT

We investigate the bargaining abilities of LLM's and how reinforcement learning (RL) can affect these abilities with a focus on the buyer role. A Group Relative Policy Optimization (GRPO) RL approach is used in combination with self-play on the Qwen2.5-7b-Instruct model in hopes to improve the sum of normalized profits (SNP) during a negotiation in a Rubenstein bargaining model. Training is performed over 651 training sessions, for 2 epochs, resulting in over 33,000 completion points used for training. As a result we notice performance increase in the buyer role for the RL trained model over 279 test sessions when evaluated against Qwen2.5-7b-Instruct Base, LLaMA-3.1-8B, and Qwen2.5-14B. During training, we also noticed a mode collapse, and discussed its origins in the GRPO training.

1 INTRODUCTION

Negotiation is a critical activity in many business focused environments that requires reasoning, strategy, understanding, and adaptation. Recent studies have tried to outline methods for evaluating the negotiation abilities of LLM's (Xia et al., 2024). In our research we use a variation of the Rubinstein bargaining model (Rubinstein, 1982) during training and testing to define a situation which encapsulates a two party turn based negotiation.

A recent study based around evaluating LLMs performance in a variation of the Rubinstein bargaining model finds that "playing buyer is more difficult that playing seller" (Xia et al., 2024). This may be due to the reason that the buyer makes the first, setting the context for negotiation, with no negotiation anchor the buyer often struggles making a beneficial first move, with (Xia et al., 2024) citing that buyer tends to start the negotiation with a bid slightly lower than their budget affecting the end deal price and inflating the seller profits. This motivates the question: can RL improve the buyer performance in a turn to turn negotiation?

We take a GRPO (DeepSeek-AI et al., 2025) self-play approach to fine-tuning the Qwen 2.5-7B-Instruct model, where the model is trained as both the buyer and seller. This approach uses a terminal-based reward model where the reward is derived from the normalized profits of the negotiation session. The model is trained and evaluated on the CamelCamel Amazon price history dataset, and evaluated using the metrics defined in (Xia et al., 2024).

2 RL METHOD AND SETUP

2.1 Bargaining Environment

We adopt the bargaining environment definition from (Xia et al., 2024), where each session involves a buyer and seller who take turns negotiating the purchase and sale, respectively, of an item. The buyer is provided a budget, B, and the seller a cost, C. The buyer initiates the negotiation, then both parties go back and forth exchanging structured utterances. Each utterance must follow a Thought, Talk, Action output format. The buyer has a set of action moves, where one must be present for a valid turn; [BUY], [DEAL], [REJECT], [QUIT]. Likewise the seller has a set of action moves, where one must be present; [SELL], [DEAL], [REJECT], [QUIT]. The negotiation continues until a deal is made, one party quits the negotiation, or both parties reach the turn limit.

Given a buyer's budget, B, and a seller's cost, C, there are two potential scenarios during a negotiation. The first being Mutual Interest, where $B > C$, this allows both parties to make a deal resulting in profit for each party. The second being Conflicting Interest, where $B <= C$, this means that at most one party can profit from a deal. In a mutual interest scenario both parties should work together to find a deal price within the range (B, C), while in a conflicting interest scenario both parties should quit.

2.2 Reward Function

For each session normalized profit (NP) is used to evaluate the buyer and seller performance. Normalized profit is the profit made by a party over the difference between B and C:

$$NPb = (B - D) / |B - C|$$

$$NPs = (D - C) / |B - C|$$

Where D is the deal price.

For negotiations that do not end in a deal, both parties are assigned a NP equal to 0. In our RL setup, NPb and NPs are used as the reward function for the buyer and seller, respectively. Each input-output pair in a session is assigned the terminal-based NP reward. A terminal reward-based RL model was chosen in aim of allowing the model to explore a wide range of strategies. The consequence of this decision is that the reward model cannot identify which individual move(s) were more important than others.

2.3 Self-Play GRPO

We implement a self-play GRPO training setup. For each product in the training split, the model plays against itself as both the buyer and seller over 8 sessions, each output and session varies as the model is queried with temperature =0.5 and top_p = 0.8.

NPb and NPs are retrieved for each session and an advantage is computed for each session, with buyer and seller advantages being computed independently. Buyer advantage is computed as ADVb = NPb_i - mean(NPb across 8 sessions) and seller advantage is computed as ADVs = NPs_i - mean(NPS across 8 sessions). Because seller and buyer advantages are computed separately a buyer advantage can be positive even though it has a negative NPb and likewise with seller roles. The policy loss uses a clipped ratio objective with $\varepsilon = 0.2$ and a KL penality with $\beta = 0.02$ against a frozen reference model.

Throughout training the model is hosted and queried on a vLLM server. Offline gradient updates are triggered after 128 reward assigned input-output pairs are accumulated across both buyers and sellers. After 4 gradient updates are performed the most recently updated model weights are reloaded into the vLLM server, and inference is now performed on the updated model.

During training a a learning rate of 1×10^{-6} (AdamW 8-bit), along with gradient clipping 1.0, batch size 8, bfloat16 precision. The model was trained over 2 epochs of the 651 product training split, which resulted in 247 gradient update checkpoints. Checkpoint 247 was used for all the model evaluation.

2.4 Mode Collapse During Training.

It was found while investigating intermediate checkpoints at one period during training that the model was experiencing mode collapse, where a pattern/trend appeared in the buyer performance. Very frequently the buyer role would experience negative profits and NPb due to accepting deals above budget, while the seller role was able to extract large profits and result in high NPs.

We believe the root cause of this is the reward and advantage model and namely the advantage being computed independently for the buyer and seller role. Because of this, the reward value is normalized by the mean over all NPb values, allowing advantage to be positive for negative NPb values. Resulting in instances where the model is reinforced for negative-profit decision making.

By the end of training this mode-collapse seemed to self-correct itself which is indicative of the final checkpoint evaluations. This correction is likely what allowed the RL to improve buyer performance.

2.5 Evaluation Metrics

The primary evaluation metrics used are deal rate, the percentage of negotiations that resulted in a deal out of all evaluation negotiations, and sum of normalized profits (SNP), where SNP = the sum of NP over all negotiation sessions.

SNP is calculated in two ways, below you will see raw and filtered SNP for both the buyer and seller. Raw SNP is the sum of NP over all negotiation sessions within the test dataset split, while filtered SNP is the sum of NP over all negotiation sessions where $|B - C| >= 1$. As the difference between the buyer budget, B, and seller cost, C approaches 0 we start to see denominator instability in NP and approach division by 0, which can greatly affect SNP. Filtered SNP provides a more stable picture of model behavior.

3 DATASET

We used the AmazonHistoryPrice dataset from the camelcamelcamel website which contains 930 datapoints, each consisting of a product title, product description, highest price, and lowest price. The buyer budget, B, is derived from the highest price as $B = 0.8 \times highest\_price$, and the seller cost, C, is derived from the lowest price as C = lowest_price. B is derived by multiplying highest_price by 0.8 to allow conflicting interest scenarios to appear which better reflects real world negotiations. This dataset contains products from across 10 product categories.

We use a 70/30 training/test split, where 651 products from the AmazonHistoryPrice are used for training and the other 279 are used for training. All evaluation results are derived from 279 sessions per model matchup.

4 RESULT

Qwen Instruct RL vs Qwen Instruct Base — performance comparison

279 sessions per matchup. SNP (sum of normalized profits) measures the total normalized profit accumulated by each side across all sessions. Raw includes all valid sessions; filtered excludes sessions where |Budget − Cost| < $1 to prevent near-zero spread distortion.

Qwen Instruct RL RL-trained model (GRPO checkpoint)   Qwen Instruct Base Base instruct model   LLaMA-3.1-8B Meta LLaMA 3.1   Qwen2.5-14B Qwen 14B instruct

Overall   Mutual interest (MI)   Conflicting interest (CI)

HEAD-TO-HEAD — QWEN INSTRUCT RL VS QWEN INSTRUCT BASE

BUYER	SELLER	DEAL RATE	RAW SNPB	RAW SNPS	FILT. SNPB	FILT. SNPS
Qwen Instruct RL buyer	Qwen Instruct Base seller	47.3%	+338.68	-228.68	+93.74	+16.26
Qwen Instruct Base buyer	Qwen Instruct RL seller	82.4%	-4558.08	+4756.08	-50.08	+246.08

Raw = all valid sessions | Filtered = sessions with |B−C| ≥ $1 | SNPb = sum of normalized profits (buyer) | SNPs = sum of normalized profits (seller) | Higher SNP = more profit captured

VS LLAMA-3.1-8B

BUYER	SELLER	DEAL RATE	RAW SNPB	RAW SNPS	FILT. SNPB	FILT. SNPS
Qwen Instruct Base buyer	LLaMA-3.1-8B seller	45.5%	-291.32	+395.32	-37.32	+141.32
Qwen Instruct RL buyer	LLaMA-3.1-8B seller	36.5%	+5106.21	-5011.21	+110.21	-16.21
LLaMA-3.1-8B buyer	Qwen Instruct Base seller	88.8%	+12685.88	-12468.88	+188.91	+27.09
LLaMA-3.1-8B buyer	Qwen Instruct RL seller	83.8%	+14935.44	-14725.44	+187.44	+19.56
RL VS BASE DIFF — POSITIVE = RL ADVANTAGE						
RL buyer − Base buyer (vs same seller)		-0.09	+5397.53	-5406.53	+147.53	-157.53
RL seller − Base seller (vs same buyer)		-0.05	+2249.56	-2256.56	-1.47	-7.53

Raw = all valid sessions | Filtered = sessions with |B−C| ≥ $1 | SNPb = sum of normalized profits (buyer) | SNPs = sum of normalized profits (seller) | Higher SNP = more profit captured

VS QWEN2.5-14B

BUYER	SELLER	DEAL RATE	RAW SNPB	RAW SNPS	FILT. SNPB	FILT. SNPS
Qwen Instruct Base buyer	Qwen2.5-14B seller	48.8%	+461.28	-348.28	-30.72	+141.72
Qwen Instruct RL buyer	Qwen2.5-14B seller	35.4%	+18.28	+71.72	+22.28	+66.72
Qwen2.5-14B buyer	Qwen Instruct Base seller	83.2%	-7568.51	+7776.51	-68.97	+275.97
Qwen2.5-14B buyer	Qwen Instruct RL seller	81.6%	-6058.56	+6264.56	-56.56	+259.56
RL VS BASE DIFF — POSITIVE = RL ADVANTAGE						
RL buyer − Base buyer (vs same seller)		-0.13	-443.00	+420.00	+53.00	-75.00
RL seller − Base seller (vs same buyer)		-0.02	+1509.95	-1511.95	+12.41	-16.41

Raw = all valid sessions | Filtered = sessions with |B−C| ≥ $1 | SNPb = sum of normalized profits (buyer) | SNPs = sum of normalized profits (seller) | Higher SNP = more profit captured

The first table reflects the performance of the base Qwen2.5-7B-Instruct (Qwen Instruct Base) model when evaluated directly against the RL trained Qwen2.5-7B-Instruct model (Qwen Instruct RL). It is important to remember the difference between raw and filtered SNP in reviewing all charts. This chart shows that the RL model is able to be profitable against the base model in both the buyer and seller role, while the base model is only able to be profitable as the seller, when looking at filtered SNP. We can conclude that Rl training improved buyer-side performance more than seller-side performance.

The second table reflects the performance of the base Qwen2.5-7B-Instruct model and Qwen2.5-7B-Instruct model when matched up against the LLaMA-3.1-8B model. To view the change in performance from the base to RL model it is important to view the base Qwen vs LLaMA evaluations compared against the RL Qwen vs LLaMA evaluations (the bottom two rows). Both the base and Rl model are strong in the seller role against the LLaMA model, with the base model outperforming the RL model by +7.53 filtered SNPs. The RL model strongly outperforms the base model in the buyer role when looking at filtered SNPb with a +147.53 filtered SNPb point increase.

The third table reflects the performance of the base Qwen2.5-7B-Instruct model and Qwen2.5-7B-Instruct model when matched up against the Qwen2.5-14B-Instruct model. Both the base and RL model struggle as the buyer, but the RL model ends with a positive filtered SNPb, and a +53.00 point increase in SNPb vs the base model. Again the base model outperforms the RL model in the seller role when matched against the Qwen2.5-14B-Instruct model with the base model outperforming the RL model by +16.41 filtered SNPs, which is a small edge compared to the large filtered SNPs that both models achieve.

5 DISCUSSION

The main finding is that the Qwen RL model performs better as the buyer which is a reverse of the typical pattern outlined in (Xia et al., 2024) that states that most models struggle as the buyer. We believe this change in buyer performance is due to the terminal NP reward in the RL model rewarding good initiating offers. It is hard to pinpoint the exact reason why this change occurred due to the terminal-based reward model allowing vast exploration during the RL training, which can be connected to the change in buyer performance throughout the training logs.

The mode-collapse observed during training highlights an important aspect of our RL and reward design. Due GRPO advantage is computed separately for the buyer and seller sessions the model was allowed to swing towards very negative buyer performance as a result of positive advantage feedback even though the model was not profitable. We note some fixes to this problem such as computing buyer and seller GRPO advantage together and including intermediate-reward values within our design instead of rolling out terminal-based reward values to each output in a negotiation session.

Near zero budget and cost differences causes a large discrepancy in raw vs filtered SNP, revealing these near-zero spreads as problematic and potentially misleading when evaluating a model's performance over multiple negotiation sessions. We advise future work using the (Xia et al., 2024) outlined bargaining method to include filtered metrics in addition to raw metric or apply a minimum budget-cost difference when curating the dataset.

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A limitation of our RL model stems from the self-play. While the goal is for both the buyer and seller to grow together and explore different strategies while allowing the other role to experience new strategies, because of self-play there is a chance that the trained model doesn't generalize well to new opponents or architectures.

6 CONCLUSION

We conclude by presenting a self-play terminal-reward based GRPO training design based on the bargaining environment outlined in (Xia et al., 2024). We observer that the resulting trained model has an increase in buyer-side performance supported by the trained model achieving an +110.21 filtered SNPb as the buyer against LLaMA-3.1-8B compared to a -37.72 filtered SNPb as the buyer against LLaMA-3.1-8B that the base model achieved, and the trained model achieving an +22.28 filtered SNPb as the buyer against Qwen2.5-14B-Instruct compared to a -30.72 filtered SNPb as the buyer against Qwen2.5-14B-Instruct that the base model achieved. This is a change in the typical trend of poor model buyer performance outlined in (Xia et al., 2024).

It is seen that terminal-based RL in self-play can strongly influence a model's bargaining behavior. We note that terminal-based reward rollouts in self play can cause mode-collapse and fail to provide opponent variation. In the future we hope to ask if adding intermediate-based rewards can help solve the faults seen in our RL setup.

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